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ECEN 474/704
Lab 6: Prelab
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1. **Comparison of differential-amplifier topologies.** Compare the two single-ended differential amplifiers in this lab. Rate differential gain, common-mode gain, output voltage swing, CMRR, and common-mode input voltage range. Include expressions for each specification.

   Solution.

   The comparison assumes comparable currents and device parameters. "Good" denotes larger differential gain, smaller common-mode gain magnitude, larger voltage range, or higher CMRR.

Specification	Simple (Fig. 6-6)	Cascode (Fig. 6-8)
Differential gain	Medium	Good
Common-mode gain ($\lvert A_{CM}\rvert$)	Medium	Good
Output voltage swing	Good	Bad
CMRR	Medium	Good
Common-mode input voltage range	Good	Bad

   With $V_{DM} = V_{i+} - V_{i-}$ and $V_{CM} = (V_{i+} + V_{i-})/2$, zero differential input gives $I_{D1} = I_{D2} = I_{\text{tail}}/2$. The overdrive voltages are $V_{OVn} = V_{GS} - V_{TN}$ and $V_{OVp} = V_{SG} - |V_{TP}|$, both positive.

   (a) Differential gain. The single-ended gain has no factor of $1/2$ because the PMOS mirror combines the two input-branch signal currents:

$$\begin{aligned} A_{DM,\text{simple}} &\approx g_{m1}(r_{o2} \parallel r_{o6}), \\ R_{P,\text{cas}} &\approx r_{o8} + r_{o10} + g_{m8} r_{o8} r_{o10}, \\ A_{DM,\text{cas}} &\approx g_{m1}(r_{o2} \parallel R_{P,\text{cas}}) \approx g_{m1} r_{o2}. \end{aligned}$$

   The cascode increases the load resistance, but M2 is not cascoded; its r_{o2} therefore limits the differential gain.

   (b) Common-mode gain. Using the low-frequency approximations in the Lab 6 handout,

$$\begin{aligned} A_{CM,\text{simple}} &\approx -\frac{1}{2 r_{o4} g_{m5}}, \\ R_{SS,\text{cas}} &\approx r_{o4} + r_{o6} + g_{m4} r_{o4} r_{o6}, \\ A_{CM,\text{cas}} &\approx -\frac{1}{2 R_{SS,\text{cas}} g_{m9}}. \end{aligned}$$

   The cascode tail source has a higher resistance and reduces $|A_{CM}|$.

   (c) Common-mode rejection ratio. For either topology,

$$\boxed{\text{CMRR} = 20 \log_{10} \left| \frac{A_{DM}}{A_{CM}} \right|.}$$

(d) Common-mode input voltage range. For the simple amplifier, the saturation constraints give

$$\begin{aligned} V_{CM,\min} &= V_{SS} + V_{OV4} + V_{TN1} + V_{OV1}, \\ V_{CM,\max} &= V_{DD} - |V_{TP5}| - V_{OV5} + V_{TN1}, \\ \Delta V_{CM} &= V_{CM,\max} - V_{CM,\min}. \end{aligned}$$

For the cascode amplifier, the voltage reserved across the lower NMOS mirror device is $d_n = V_{S4} - V_{SS}$. With a fixed V_{BN},

$$\begin{aligned} d_n &= V_{BN} - V_{SS} - V_{TN4} - V_{OV4}, \\ V_{CM,\min} &= V_{SS} + d_n + V_{OV4} + V_{TN1} + V_{OV1}, \\ V_{CM,\max} &= V_{DD} - |V_{TP9}| - V_{OV9} + V_{TN1}. \end{aligned}$$

The handout's minimum-headroom expression is the special case $d_n = V_{OV6}$. A practical choice $d_n > V_{OV6}$ provides saturation margin but raises the lower input limit. These ranges also require the reference branches and output devices to remain in saturation; output bias can impose an additional input limit.

(e) Output voltage swing. Let V_S be the common source voltage of M1 and M2. At a fixed common-mode input and with matched input devices,

$$V_{o,\min} = V_S + V_{OV2} = V_{CM} - V_{TN1}.$$

For the simple circuit, the upper output limit is $V_{o,\max} = V_{DD} - V_{OV6}$. For the cascode circuit, $d_p = V_{DD} - V_{S8}$ gives

$$d_p = V_{DD} - V_{BP} - |V_{TP8}| - V_{OV8}, \qquad V_{o,\max} = V_{DD} - d_p - V_{OV8}.$$

The ideal choice $d_p = V_{OV10}$ gives $V_{o,\max} = V_{DD} - V_{OV10} - V_{OV8}$. In both cases the available interval width is $\Delta V_o = V_{o,\max} - V_{o,\min}$, while the largest symmetric signal about $V_{o,Q}$ has peak amplitude

$$v_{o,\mathrm{pk}} < \min(V_{o,Q} - V_{o,\min},\, V_{o,\max} - V_{o,Q}).$$

These are saturation-based bounds; a specified distortion limit would require additional margin inside the interval.

Parameters and assumptions for Questions 2 and 3. The single-finger $W/L = 5/0.27\,\mu$m data in Table I of the Lab 1 report provide the device parameters:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ (μA/V^2)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All initial lengths are $0.27\,\mu$m. The DC sizing uses the square-law model and neglects body effect, mobility degradation, and channel-length-modulation corrections to the drain current. The parameter λ enters only the small-signal estimate $r_o \approx 1/(\lambda I_D)$. All widths are total effective widths. The resulting dimensions constitute an analytical starting point.

2. **Simple differential-amplifier design (Fig. 6-6).** Design for SR $> 15\,\mathrm{V}/\mu\mathrm{s}$, GBW $> 8\,\mathrm{MHz}$, and $\Delta V_{CM} > 0.4\,\mathrm{V}$, with $V_{DD} = 0.9\,\mathrm{V}$, $V_{SS} = -0.9\,\mathrm{V}$, and $C_L = 10\,\mathrm{pF}$.

Solution.

(a) Tail current and input transconductance. The slew-rate constraint requires

$$I_{\mathrm{tail}} > C_L(15 \times 10^6) = 150\,\mu\mathrm{A}.$$

The handout gives angular bandwidth $\omega_{GBW} = g_{m1}/C_L$. Converting to hertz gives

$$\mathrm{GBW} = \frac{g_{m1}}{2\pi C_L}, \qquad g_{m1} > 2\pi(8 \times 10^6)(10 \times 10^{-12}) = 0.503\,\mathrm{mS}.$$

For $I_{\mathrm{tail}} = I_{\mathrm{bias}} = 180\,\mu\mathrm{A}$ and a 1:1 M3–M4 mirror, each input and PMOS load device carries $90\,\mu\mathrm{A}$ at the balanced operating point. The selected overdrives are

$$V_{OV1,2} = 0.250\,\mathrm{V}, \quad V_{OV3,4} = 0.150\,\mathrm{V}, \quad V_{OV5,6} = 0.200\,\mathrm{V}.$$

Then $g_{m1} = 2(90\,\mu\mathrm{A})/(0.250\,\mathrm{V}) = 0.720\,\mathrm{mS}$, providing margin above the required transconductance.

(b) Transistor widths and lengths. The square-law sizing relations are

$$I_D \approx \frac{1}{2}K_P\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2I_D}{K_P V_{OV}^2}, \qquad W = L\frac{W}{L}.$$

The sizing calculations, with currents in $\mu\mathrm{A}$, are

$$(W/L)_{1,2} = \frac{2(90)}{215.6(0.250)^2} = 13.358,$$
$$(W/L)_{3,4} = \frac{2(180)}{215.6(0.150)^2} = 74.212,$$
$$(W/L)_{5,6} = \frac{2(90)}{47.57(0.200)^2} = 94.597.$$

Device	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2, each	90	0.250	3.607	0.270
M3, M4, each	180	0.150	20.037	0.270
M5, M6, each	90	0.200	25.541	0.270

NMOS bodies connect to V_{SS} and PMOS bodies to V_{DD}. Matched device pairs use equal dimensions and consistent finger configurations. The diode-connected M3 establishes the common gate voltage, eliminating the need for a separate tail-bias voltage source.

(c) Common-mode range. Substituting the selected overdrives gives

$$V_{CM,\mathrm{min}} = -0.9 + 0.150 + 0.433 + 0.250 = -0.067\,\mathrm{V},$$
$$V_{CM,\mathrm{max}} = 0.9 - 0.4068 - 0.200 + 0.433 = 0.7262\,\mathrm{V},$$
$$\Delta V_{CM} = \boxed{0.7932\,\mathrm{V} > 0.4\,\mathrm{V}.}$$

The selected bias $V_{CM,Q} = 0.300\,\mathrm{V}$ lies inside this interval.

(d) Nominal DC operating point and saturation checks. For equal inputs, a symmetry-consistent operating point places the output at the diode-connected PMOS voltage:

$$\begin{aligned} V_S &= 0.300 - (0.433 + 0.250) = -0.3830\,\text{V}, \\ V_{D1} = V_{o,Q} &= 0.900 - (0.4068 + 0.200) = 0.2932\,\text{V}, \\ V_{G3} = V_{G4} &= -0.900 + 0.433 + 0.150 = -0.3170\,\text{V}. \end{aligned}$$

Each device satisfies $V_{DS} > V_{OV}$ for an NMOS or $V_{SD} > V_{OV}$ for a PMOS under the assumed model:

Device	V_{DS} or V_{SD} (V)	V_{OV} (V)	Saturation
M1, M2	0.6762	0.250	Yes
M3	0.5830	0.150	Yes
M4	0.5170	0.150	Yes
M5, M6	0.6068	0.200	Yes

At $V_{CM,Q} = 0.300\,\text{V}$, the output limits are

$$-0.133\,\text{V} < V_o < 0.700\,\text{V}.$$

The interval is 0.833 V wide. About $V_{o,Q} = 0.2932\,\text{V}$, the symmetric-swing limit approaches 0.4068 V peak. A finite saturation margin reduces the usable amplitude below this limit.

(e) Small-signal performance. Using the stated λ values,

$$\begin{aligned} r_{o2} &= \frac{1}{0.1624(90\times10^{-6})} = 68.418\,\text{k}\Omega, \\ r_{o6} &= \frac{1}{0.1945(90\times10^{-6})} = 57.127\,\text{k}\Omega, \\ r_{o4} &= \frac{1}{0.1624(180\times10^{-6})} = 34.209\,\text{k}\Omega, \\ R_{\text{out}} &= r_{o2} \parallel r_{o6} = 31.132\,\text{k}\Omega. \end{aligned}$$

Thus $A_{DM} = g_{m1}R_{\text{out}} = 22.415\,\text{V/V}$ (27.01 dB). With $g_{m5} = 2(90\,\mu\text{A})/(0.200\,\text{V}) = 0.900\,\text{mS}$,

$$A_{CM} \approx -\frac{1}{2r_{o4}g_{m5}} = -0.01624, \qquad \text{CMRR} \approx 62.80\,\text{dB}.$$

The dominant-pole approximation gives

$$f_{p1} = \frac{1}{2\pi R_{\text{out}}C_L} = 511.22\,\text{kHz}, \quad \text{GBW} = \frac{g_{m1}}{2\pi C_L} = 11.46\,\text{MHz}.$$

Specification	Requirement	Analytical estimate
Slew rate, I_{tail}/C_L	$> 15\,\text{V}/\mu\text{s}$	$18\,\text{V}/\mu\text{s}$
GBW	$> 8\,\text{MHz}$	11.46 MHz
Common-mode range width	$> 0.4\,\text{V}$	0.7932 V

3. **Differential amplifier with cascode mirrors (Fig. 6-8).** Design for SR $> 7\,\mathrm{V}/\mu\mathrm{s}$, GBW $> 8\,\mathrm{MHz}$, and CMRR > 65 dB, with $V_{DD} = 0.9\,\mathrm{V}$, $V_{SS} = -0.9\,\mathrm{V}$, and $C_L = 10\,\mathrm{pF}$. Determine V_{BP} and V_{BN} and retain a nonzero input common-mode range.

Solution.

(a) Current, overdrives, and transistor dimensions. The minimum tail current is $C_L(7 \times 10^6) = 70\,\mu\mathrm{A}$; the input-transconductance minimum remains 0.503 mS. A 1:1 tail mirror establishes

$$I_{\mathrm{bias}} = I_{\mathrm{tail}} = 120\,\mu\mathrm{A}, \qquad I_{D1,2,7,8,9,10} = 60\,\mu\mathrm{A}.$$

For $V_{OV} = 0.150\,\mathrm{V}$ in all ten devices and $L = 0.270\,\mu\mathrm{m}$, the sizing equation of Question 2 gives

$$(W/L)_{1,2} = \frac{2(60)}{215.6(0.150)^2} = 24.737,$$
$$(W/L)_{3,4,5,6} = \frac{2(120)}{215.6(0.150)^2} = 49.474,$$
$$(W/L)_{7,8,9,10} = \frac{2(60)}{47.57(0.150)^2} = 112.115.$$

Device	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2, each	60	0.150	6.679	0.270
M3–M6, each	120	0.150	13.358	0.270
M7–M10, each	60	0.150	30.271	0.270

The input transconductance is $g_{m1} = 2(60\,\mu\mathrm{A})/(0.150\,\mathrm{V}) = 0.800\,\mathrm{mS}$.

(b) Cascode bias voltages. Reserving $d_n = d_p = 0.200\,\mathrm{V}$ across the lower NMOS and upper PMOS mirror devices gives 50 mV of nominal saturation margin above their 0.150 V overdrives:

$$V_N = V_{S3} = V_{S4} = V_{SS} + d_n = -0.700\,\mathrm{V},$$
$$V_P = V_{S7} = V_{S8} = V_{DD} - d_p = 0.700\,\mathrm{V},$$
$$V_{BN} = V_N + V_{TN4} + V_{OV4} = \boxed{-0.117\,\mathrm{V},}$$
$$V_{BP} = V_P - |V_{TP8}| - V_{OV8} = \boxed{0.1432\,\mathrm{V}.}$$

As permitted by the handout, external sources at the testbench supply these biases through additional schematic pins. The amplifier therefore requires no internal bias-generation transistors or additional bias current for these voltages.

(c) Common-mode input range with these biases. Including the selected d_n, rather than only its ideal minimum,

$$V_{CM,\mathrm{min}} = -0.900 + 0.200 + 0.150 + 0.433 + 0.150 = 0.0330\,\mathrm{V},$$
$$V_{CM,\mathrm{max}} = 0.900 - 0.4068 - 0.150 + 0.433 = 0.7762\,\mathrm{V}.$$

Hence $\boxed{\Delta V_{CM} = 0.7432\,\mathrm{V} > 0}$. The nominal operating point is $V_{CM,Q} = 0.300\,\mathrm{V}$.

(d) DC operating point and saturation checks. In Fig. 6-8, M3's drain drives the gates of M5 and M6; M7's drain drives the gates of M9 and M10. Therefore,

$$V_{G5} = V_{G6} = V_{D3} = -0.900 + 0.433 + 0.150 = -0.3170\,\text{V},$$
$$V_{D1} = V_{G9} = V_{G10} = V_{o,Q} = 0.900 - 0.4068 - 0.150 = 0.3432\,\text{V},$$
$$V_S = 0.300 - 0.433 - 0.150 = -0.2830\,\text{V}.$$

Together with $V_N = -0.700\,\text{V}$ and $V_P = 0.700\,\text{V}$, these voltages give the following first-order saturation checks:

Device	V_{DS} or V_{SD} (V)	V_{OV} (V)	Saturation
M1, M2	0.6262	0.150	Yes
M3	0.3830	0.150	Yes
M4	0.4170	0.150	Yes
M5, M6	0.2000	0.150	Yes
M7, M8	0.3568	0.150	Yes
M9, M10	0.2000	0.150	Yes

At the chosen common-mode voltage, the output interval is $-0.133\,\text{V} < V_o < 0.550\,\text{V}$. Its width is 0.683 V, and the symmetric swing about $V_{o,Q} = 0.3432\,\text{V}$ approaches 0.2068 V peak. The smaller upper headroom is the cost of the cascode PMOS load.

(e) Gain, common-mode rejection, and frequency estimates. The input and PMOS transconductances are 0.800 mS; the NMOS tail-device transconductances are 1.600 mS. Using $r_o \approx 1/(\lambda I_D)$ gives

$$r_{o2} = 102.627\,\text{k}\Omega, \quad r_{o4} = r_{o6} = 51.314\,\text{k}\Omega, \quad r_{o8} = r_{o10} = 85.690\,\text{k}\Omega.$$

Consequently,

$$R_{SS} \approx r_{o4} + r_{o6} + g_{m4} r_{o4} r_{o6} = 4.3156\,\text{M}\Omega,$$
$$R_P \approx r_{o8} + r_{o10} + g_{m8} r_{o8} r_{o10} = 6.0456\,\text{M}\Omega,$$
$$R_{\text{out}} \approx r_{o2} \parallel R_P = 100.914\,\text{k}\Omega,$$
$$A_{DM} \approx g_{m1} R_{\text{out}} = 80.731\ \text{V/V} \quad (38.14\,\text{dB}),$$
$$A_{CM} \approx -\frac{1}{2 R_{SS} g_{m9}} = -1.4482 \times 10^{-4},$$
$$\text{CMRR} \approx 20 \log_{10} \left| \frac{80.731}{-1.4482 \times 10^{-4}} \right| = 114.92\,\text{dB}.$$

Specification	Requirement	Analytical estimate
Slew rate, I_{tail}/C_L	$> 7\,\text{V}/\mu\text{s}$	$12\,\text{V}/\mu\text{s}$
GBW, $g_{m1}/(2\pi C_L)$	$> 8\,\text{MHz}$	12.73 MHz
CMRR	> 65 dB	114.92 dB
Common-mode range width	Nonzero	0.7432 V
Dominant pole, $1/(2\pi R_{\text{out}} C_L)$	Record estimate	157.71 kHz

Model validity and comparison with simulation results.
The calculated dimensions satisfy the requirements within the stated first-order model. The Lab 1 parameters were extracted at one geometry and bias condition. Body effect, mobility degradation, finite drain-voltage-dependent currents, internal poles, and parasitic capacitances are not fully represented by these estimates. Pole and GBW calculations assume the 10 pF load dominates the output capacitance and higher-frequency internal poles are negligible. The Lab 1 report provides θ but no numerical body-effect parameters. Threshold shifts caused by the elevated NMOS source voltages and PMOS cascode source voltages therefore remain unquantified. The computed CMRR assumes matched devices and ideal AC-grounded cascode biases, excluding mismatch effects.
The Lab 5 report illustrates the difference between analytical sizing and simulation-refined dimensions. Its simulated $120\,\mu$A simple mirror uses $W = 19.4\,\mu$m, $L = 0.270\,\mu$m for approximately 0.150 V compliance, whereas the first-order square-law calculation gives $W = 13.36\,\mu$m. The final cascode mirror uses $L = 0.320\,\mu$m to increase output resistance. These dimensions characterize the refined Lab 5 circuits; they do not establish a new device-parameter set for the Lab 6 designs. The present design retains $L = 0.270\,\mu$m to use the corresponding Lab 1 parameter set without assigning an unmeasured λ to $0.320\,\mu$m devices.
A common-mode DC sweep determines the interval in which all active devices remain in saturation. The widths, reference current, and V_{BN}/V_{BP} control the bias point and provide the variables for design refinement. Separate differential-mode and common-mode AC excitations determine the corresponding gains, CMRR, GBW, and dominant pole. Slew rate requires a large-signal transient analysis because it lies outside the linearized AC model. Saturation and performance margins account for the effects omitted from the first-order design.

References.

[1] S. Palermo, *ECEN 474/704 Lab 6: Differential Pairs*, course laboratory handout, Figs. 6-6 and 6-8 and Prelab Questions 1–3.

[2] Anonymous contributor, *ECEN 704 Lab 1 Report*, Table I: single-finger $5/0.27\,\mu$m NFET and PFET parameter rows.

[3] Anonymous contributor, *ECEN 704 Lab 5 Report*, Sec. III and Table 3: final mirror sizes and simulation-based refinement.